Algorithms

PATHS & CYCLES

Last Week

- Definitions
- Topological Sort

This Week

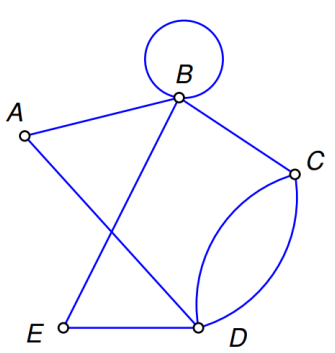
- Euler Paths & Cycles
- De Bruijn Sequence
- Hamiltonian Paths & Cycles

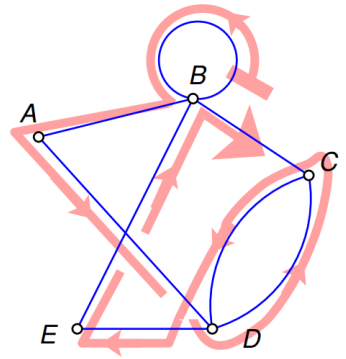
Euler Path

- An Euler path of a connected (directed/undirected) graph G= (V,E) is a path that traverses each *edge* of G exactly once.

  Notes:

  - An Euler path may visit a vertex more than once.
  - G may not be simple.





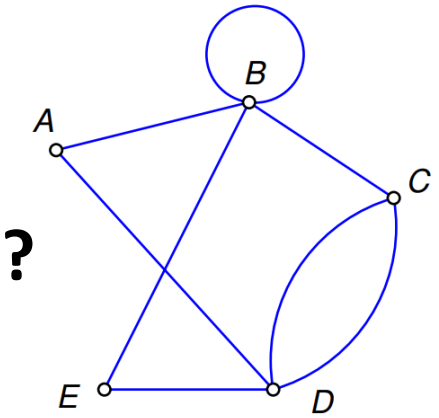
Euler Cycle

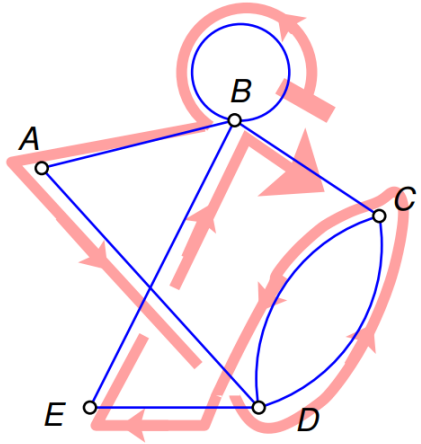
- An Euler cycle is an Euler path that starts and ends on the same vertex

Condition: all degrees are even

Is there an Euler cycle in this graph?

No. deg(C), deg(B) are odd

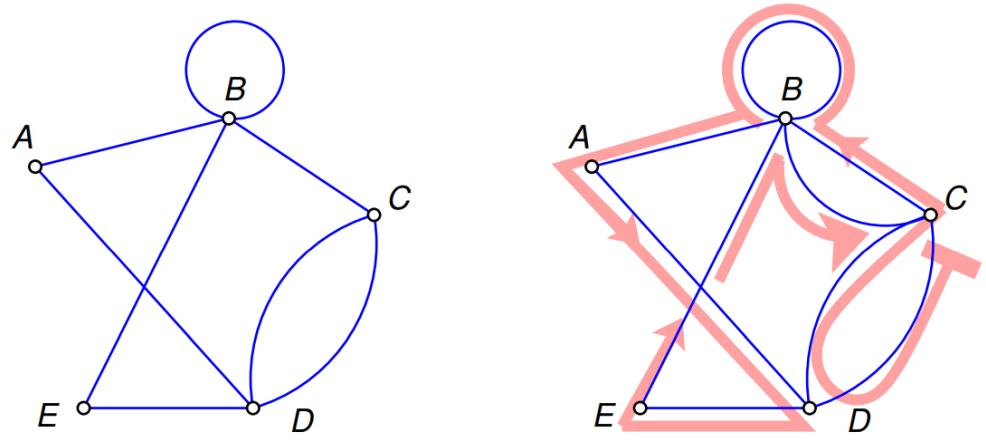




Euler Cycle

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Condition: all degrees are even



Euler Cycle

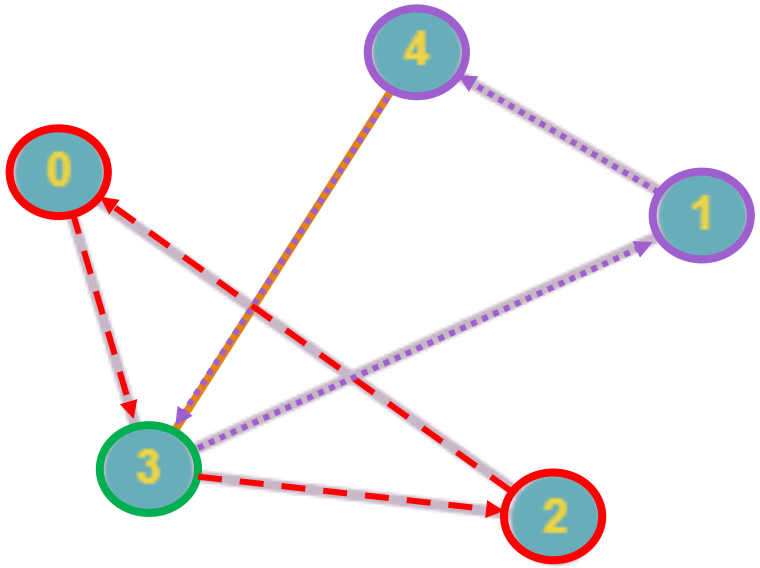
- Finding Euler cycle:

Algorithm 1 EULER CYCLE(G)

```
1: v ← Pick a vertex at random
2: Traverse the graph until returning to v
3: while There are unused edges that are adjacent to vertex in the current tour do
4:     Pick a vertex u incident to an unused edge
5:     Traverse the graph using unused edges until returning to u
6:     Join the paths
```

Euler Cycle

- Finding Euler cycle:

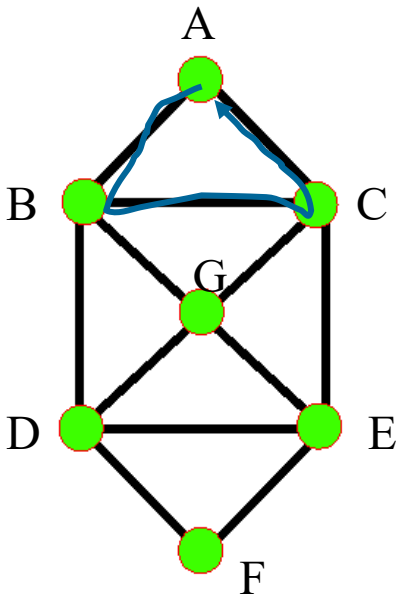


Complexity? **O(m+n)**

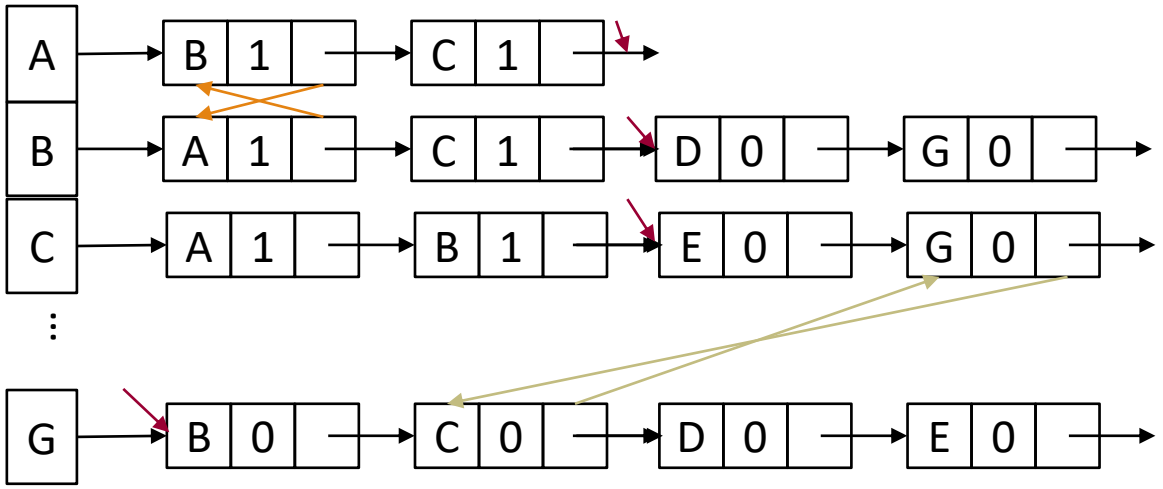
Implementation in O(|V|+|E|)

Assume the graph is represented by adj. lists.
We should keep track of already-traced edges. Can be done by adding a 0/1 entry in the corresponding link-list node.
Also add pairs of pointers  between pairs of ‘same edge’ entries + pointer ↘ to next not-traced edge.

Implementation in O(|V|+|E|)



The status after A-B-C-A is traced:



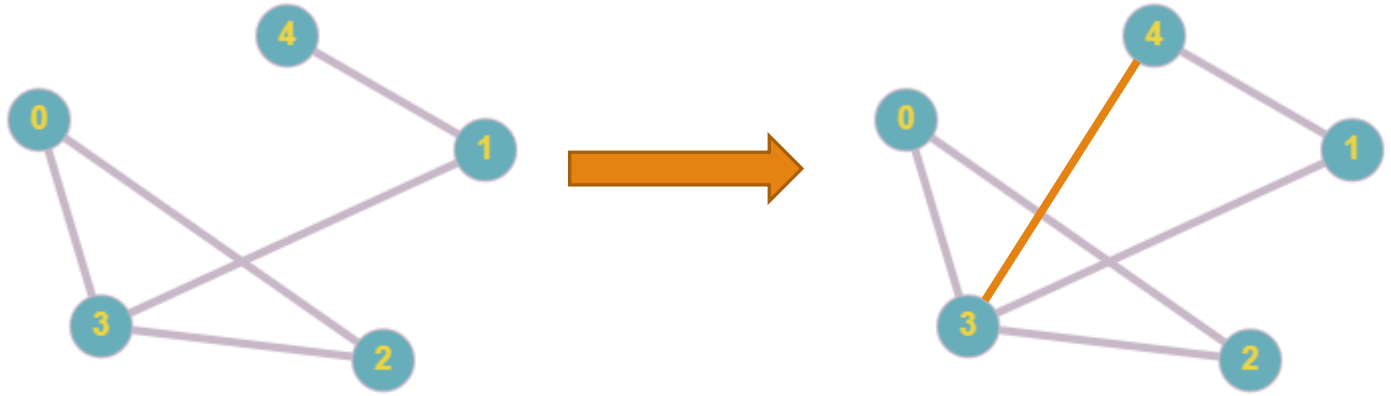
Every list is scanned exactly once along the whole algorithm.

Exercise 1

Let $G = (V, E)$ be an undirected connected graph where exactly $l = 2k, k \in \mathbb{N}$ of its vertices have an odd degree. Prove that there exists a partition of the graph edges to k edge-disjoint paths.

Exercise 1 - Solution

- Want to use Euler cycle. For that, add k edges to the graph in order to get a graph with all even degrees.
- Example for k=1:



Exercise 1 - Solution

Let $\{v_1, \dots, v_{2k}\}$ be the vertices with an odd degree. Add k new edges to the graph as follows:
$\{(v_1, v_2), (v_3, v_4) \dots, (v_{2k-1}, v_{2k})\}$.

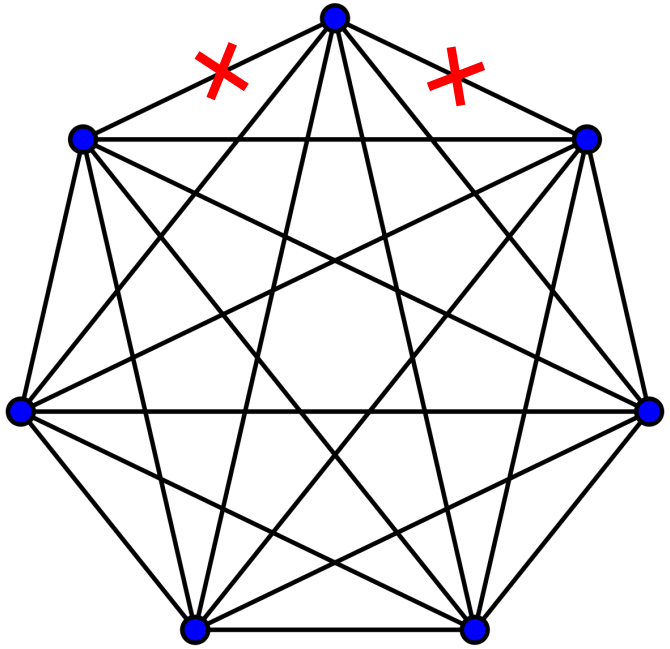
Now, all vertices have an even degree, and the resulting graph contains an Euler cycle. After removing the k added edges from the Euler cycle, we are left with k disjoint paths.

Theorems

- A connected undirected graph has an Euler cycle if and only if every vertex has an even degree
- A strongly connected directed graph has an Euler cycle if and only if for every $v \in V, d_{in}(v) = d_{out}(v)$
- A connected graph has an Euler path that is not an Euler cycle if and only if exactly two vertices have an odd degree

Exercise 2

Let G be a complete graph with 183 vertices. Two edges with a common vertex were removed from the graph. Which of the following claims are true?



Exercise 2

Let G be a complete graph with 183 vertices. Two edges with a common vertex were removed from the graph. Which of the following claims are true?

1. G contains an Euler cycle.
2. G contains an Euler path, but no Euler cycle.
3. There is no Euler cycle or Euler path in G.

Before removing – every node has degree 182

After removing – 1 node has degree 180, 2 nodes have degree 181, rest 182

Exercise 2

Let G be a complete graph with 183 vertices. Two edges with a common vertex were removed from the graph. Which of the following claims are true?

1. G contains an Euler cycle. **No – 2 vertices with odd deg.**
2. G contains an Euler path, but no Euler cycle. **Yes.**
3. There is no Euler cycle or Euler path in G.

Before removing – every node has degree 182

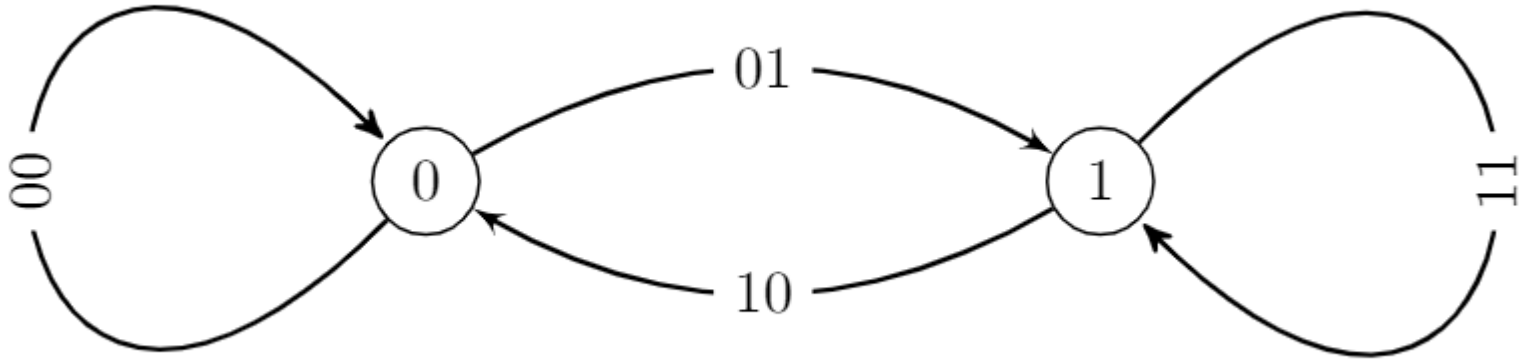
After removing – 1 node has degree 180, 2 nodes have degree 181, rest 182

De Bruijn Sequence

A De Bruijn sequence of order n on a size σ alphabet Σ is a cyclic sequence in which every possible length n string on Σ occurs exactly once as a substring

De Bruijn Sequence - Example

For $\Sigma = \{0,1\}, n = 2, \sigma = 2$: the De Bruijn Sequence is 0011.



De Bruijn Graphs - $G_{\sigma,n}$

1. $|V| = \sigma^{n-1}$
2. $|E| = \sigma^{n}$
3. The edge $b_1 b_2 \dots b_n$ starts from the vertex $b_1 b_2 \dots b_{n-1}$ and ends at the vertex $b_2 b_3 \dots b_n$
4. Every vertex has exactly σ incoming edges and σ outgoing edges
5. The graph $G_{\sigma,n}$ contains a directed Euler cycle
6. An Euler cycle corresponds to a De Bruijn sequence

Exercise 3

012012 is a De Bruijn sub-sequence over some Σ,n. Which of the statements below is true:

1. If n= 5 then in the corresponding $G_{\sigma,n}$ there must be an edge from vertex 0120 to vertex 2012.

Exercise 3 - Solution

012012 is a De Bruijn sub-sequence over some Σ,n. Which of the statements below is true:

1. If n= 5 then in the corresponding $G_{\sigma,n}$ there must be an edge from vertex 0120 to vertex 2012.

All vertices connected from 0120 are 120x, where $x \in \Sigma$.

False

Exercise 3

012012 is a De Bruijn sub-sequence over some Σ,n. Which of the statements below is true:

2. If the next character in the sequence is 0 then it must be that n > 4.

Exercise 3 - Solution

012012 is a De Bruijn sub-sequence over some Σ,n. Which of the statements below is true:

2. If the next character in the sequence is 0 then it must be that n > 4.

The sub-sequence: 0120120

For n=2,3,4: we get a repetition of a word

For n > 4 it is possible to avoid a repetition: 0120120x by choosing $x \neq 1$.

True

Exercise 3

012012 is a De Bruijn sub-sequence over some Σ,n. Which of the statements below is true:

3. If n= 4 and Σ={0,1,2,3} then the next character in the sequence must be 3.

Exercise 3 - Solution

012012 is a De Bruijn sub-sequence over some Σ,n. Which of the statements below is true:

3. If n = 4 and Σ={0,1,2,3} then the next character in the sequence must be 3.

We can check for n = 4:

0120120 – we get a repetition

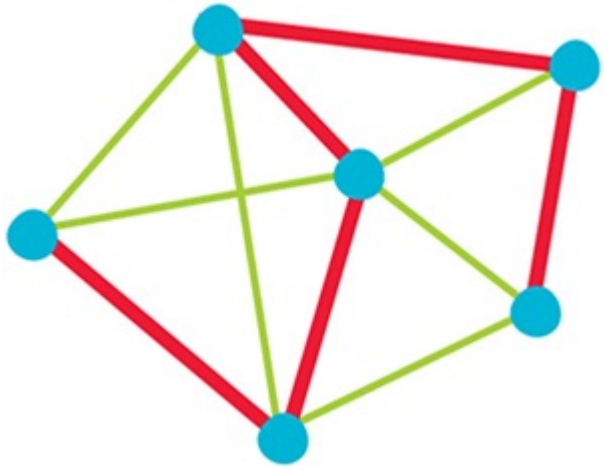
0120121 – no repetition

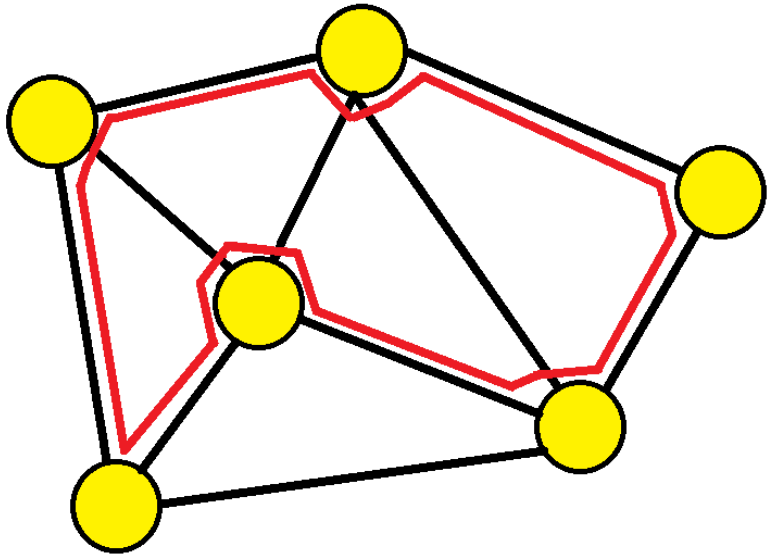
(0120122 also good, but it's enough to find 1 option)

False

Hamiltonian Path

- A Hamiltonian path of a connected graph G is a path that visits each *vertex* exactly once
- A Hamiltonian cycle of a connected graph G is a Hamiltonian path that is a cycle



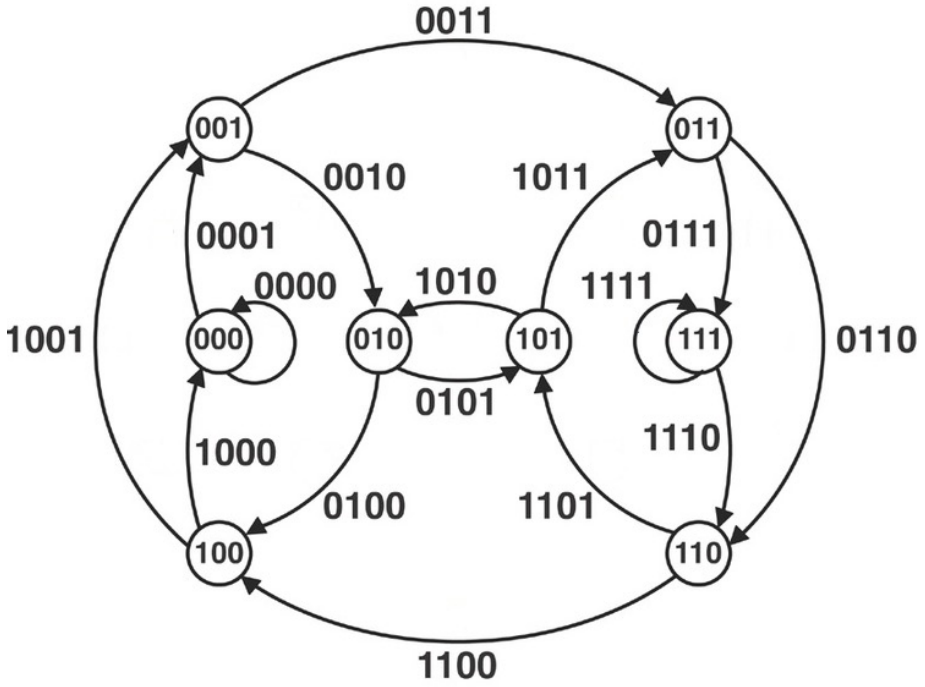


Hamiltonian Path Complexity

- Much harder than Euler path.
- Brute force algorithm runs in time O(n!), and best known algorithm runs in time O(n^2 2^n).

Theorem

For every $n \geq 2$, the graph $G_{2,n}$ has a **Hamilton** cycle.



Theorem

For every $n \geq 2$, $G_{2,n}$ has a **Hamilton** cycle.

Proof. As seen in class (property 5), we know that $G_{2,n-1}$ contains an Euler cycle. We denote that cycle:

$$C = e_1, e_2, \dots, e_{2^{n-1}}, e_1$$

Every edge in the cycle is a n-1 length word over the alphabet. Thus, every edge corresponds to a vertex in the graph $G_{2,n}$.

Explicitly, the path $C' = v_1, v_2, \dots, v_{2^{n-1}}, v_1$ exists in $G_{2,n}$, and as such is a Hamiltonian cycle.

Theorem

The length of C is $2^{n-1} = \left|V_{2,n}\right|$

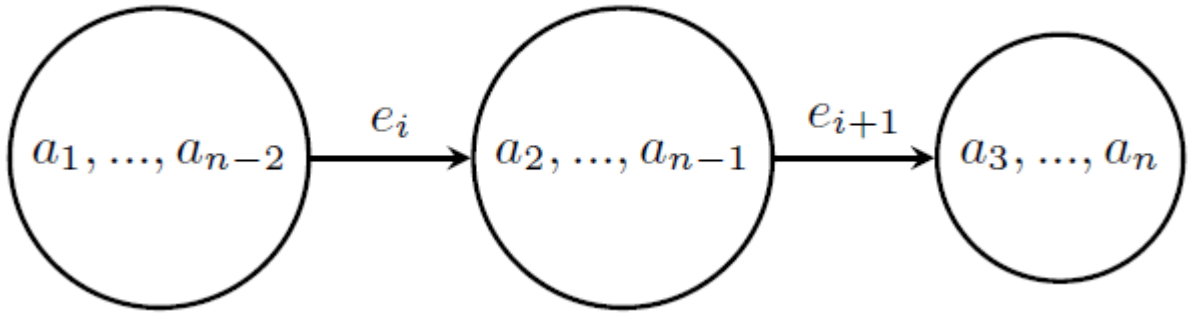
Every edge in C appears once, Thus each vertex in C' appears at most once.

Goal: Show that there is an edge between every two vertices in C'.

Theorem

Proof (continued).

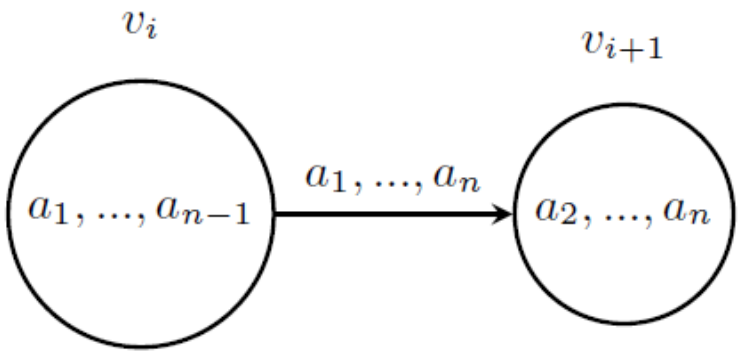
Let $e_i = a_1, \ldots, a_{n-1}$ and $e_{i+1} = a_2, \ldots, a_n$ be two edges in $G_{2,n-1}$ as follows:



Theorem

Proof (continued).

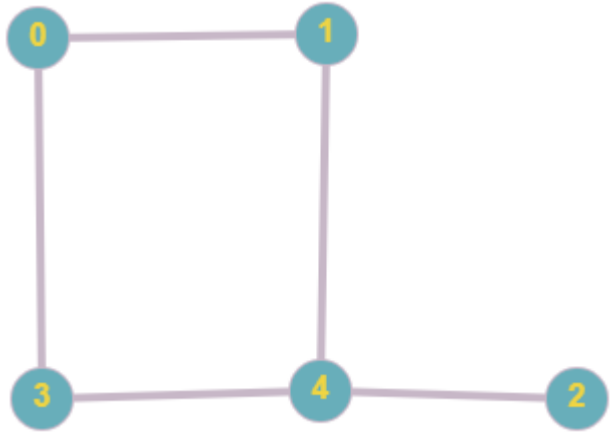
The nodes v_i and v_{i+1} are incident in $G_{2,n}$:



Exercise 4

Give an example of the following undirected graphs:

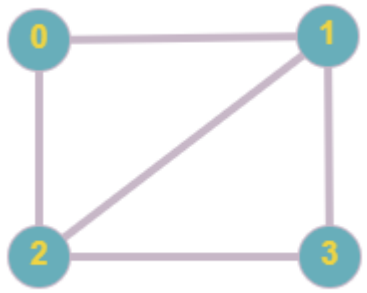
1. Graph with No Hamiltonian cycle and no Euler cycle.

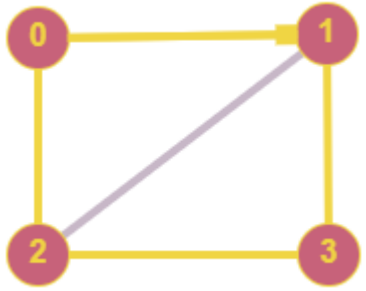


Exercise 4

Give an example of the following undirected graphs:

2. Graph with Hamiltonian cycle but no Euler cycle.

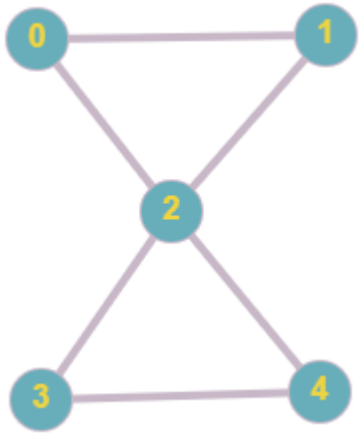




Exercise 4

Give an example of the following undirected graphs:

3. Graph with no Hamiltonian cycle but with Euler cycle.



Exercise 4

Give an example of the following undirected graphs:

4. Graph with both Hamiltonian cycle and Euler cycle.

